Practice Set 5 Paper 4 – Recent Advances

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Practice paper compiled from the question bank.

Q1. Describe energy sources for braking stones in uro

Type: LAQ (25 marks) (also answered in Practice Set 1 Paper 4, Q2)

Read as "energy sources for breaking stones in urology".

Introduction

- Stones are **brittle**: they crack under tension and shear. Tissue is **elastic** and survives the same stress. Every lithotripter exploits this difference.
- Energy reaches the stone **extracorporeally** (SWL, shock wave lithotripsy) or **intracorporeally** through an endoscope.

Classification

Group	Sources
Extracorporeal	SWL: electrohydraulic, electromagnetic, piezoelectric; BWL (burst wave lithotripsy, investigational)
Intracorporeal mechanical	Ballistic (pneumatic), ultrasonic, combined ultrasonic–pneumatic
Intracorporeal electrical	EHL (electrohydraulic lithotripsy)
Intracorporeal light	Lasers: Ho:YAG (holmium:yttrium-aluminium-garnet), TFL (thulium fibre laser), pulsed Tm:YAG (thulium:YAG); historic pulsed dye and FREDDY

1. Shock wave lithotripsy

- **Shock wave:** peak positive pressure 30–100 MPa, then a tensile tail of about −10 MPa. It is generated at F1, focused, and delivered to the stone at F2.

Generator	Mechanism	Strength	Weakness
Electrohydraulic (original HM3, 1980)	Underwater spark at F1 of an ellipsoid reflector	Very effective	Electrode life <40,000 shocks; focus drifts
Electromagnetic (modern standard)	Coil pushes a metal membrane; lens or reflector focuses	Consistent; high energy at F2; long life	—
Piezoelectric	Many ceramic elements on a spherical dish	Long life; little pain	Small focal zone, low energy

- **Fracture mechanisms:** shear and tear forces; **spallation** at the distal face; quasistatic squeezing; **cavitation** (bubble collapse erodes the surface); dynamic squeezing (most powerful). A focal zone wider than the stone works best.
- **Technique:** 60–90 shocks/min beats 120/min, because bubble clouds shield the stone and injure the kidney; voltage ramping; bubble-free gel coupling.
- Poor results: calcium oxalate monohydrate, brushite, cystine, density >1000 HU (Hounsfield units), skin-to-stone distance >10 cm.

2. Ballistic (pneumatic) lithotripsy

- Compressed air drives a projectile against a metal probe, a **jackhammer** (about 12 Hz).
- Breaks all stones cheaply with little ureteric injury. Drawbacks: **retropulsion**; rigid probes only, as flexible probes block deflection.

3. Ultrasonic lithotripsy

- A piezoceramic transducer vibrates a hollow probe at ultrasonic frequency; contact drills the stone.
- **Suction** through the hollow probe removes fragments at once. Best for soft stones in PCNL (percutaneous nephrolithotomy); weak on hard stones; needs irrigation for cooling.

4. Combined devices

- Ultrasonic plus ballistic in one probe (for example Trilogy, ShockPulse-SE): ballistic cracks hard stones while ultrasound and suction clear fragments. Faster PCNL.

5. Electrohydraulic lithotripsy

- A spark at the electrode tip forms a **vapour bubble**; expansion and collapse send pressure waves in all directions.
- Cheap; 3–5 Fr flexible probes. Poorly focused, so it injures the ureter and the scope; weak on cystine. AUA (American Urological Association) advises against it as first-line ureteric lithotripsy; still used for bladder stones.

6. Lasers

- **Pulse duration decides mechanism.** Short pulses (<10 µs: pulsed dye, FREDDY) are photomechanical, with cavitation shocks. Long pulses (>250 µs: holmium, thulium) are **photothermal**: the stone is heated until it melts, decomposes or flashes its water to steam.

	Ho:YAG	TFL
Wavelength	**2120 nm** (often quoted 2100)	1940 nm (water absorption peak)
Frequency	Up to about 120 Hz	Up to about 2000 Hz
Peak power	High	Low → less retropulsion
Fibre	≥200 µm	50–150 µm (more flow and deflection)
Product	Fragments and dust	Finer dust

- **Holmium:** standard since the 1990s; breaks **every** composition; flexible quartz fibres. Fragmenting 0.6–1.0 J × 6–10 Hz; dusting 0.2–0.3 J × 40–80 Hz; "popcorning" in a calyx.
- **Moses effect:** the start of the pulse boils a vapour channel through which the rest reaches the stone; pulse modulation reduces retropulsion.
- **TFL RCTs (randomised controlled trials):** superior to low-power (30 W) holmium; equal stone-free rate to 120 W pulse-modulated holmium.
- **Retropulsion** rises with pulse energy, not frequency, and falls with longer pulses.

Safety

- Laser heat follows total power: ≤10 W in the ureter, pauses, high flow, access sheath. Eyewear for all.
- Never fire without seeing the fibre tip on the stone.
- SWL: haematoma, arrhythmia, steinstrasse. Contraindications: pregnancy, coagulopathy or anticoagulants, untreated infection, aneurysm, distal obstruction.

Choice by site

Site	Usual energy
Ureter	Laser (semirigid or flexible); pneumatic distally; SWL
Kidney	SWL; laser via flexible URS (ureteroscopy); PCNL with ultrasonic, combined or laser
Bladder	Laser, pneumatic, EHL, mechanical

Recent advances

- **BWL:** handheld ultrasound bursts (5–100 cycles) break stones without anaesthesia. **Ultrasonic propulsion** moves fragments.
- Suction sheaths; AI (artificial intelligence)-adjusted laser settings.

Diagram to draw: SWL electrohydraulic ellipsoid with F1 and F2, stone at F2.

Key points to remember:

- SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
- Holmium 2120 nm, photothermal, breaks every composition.
- TFL 1940 nm: low peak power, finer dust, less retropulsion.
- EHL is poorly focused, so it is not first-line in the ureter.

Q2. Discuss advances in management of CA prostate

Type: LAQ (25 marks) (related: 2025-Winter Paper 3, Q3b; 2024-Winter Paper 4, Q3f)

Introduction

- Three aims drive recent change: **find significant cancer and avoid diagnosing insignificant cancer**; **match local treatment to risk** with less morbidity; give **earlier, intensified and targeted** systemic therapy.

1. Screening and diagnosis

- **ERSPC (European Randomized Study of Screening for Prostate Cancer)** and Göteborg-1: PSA (prostate-specific antigen) screening lowers cancer-specific mortality, but PSA alone causes negative biopsies and overdiagnosis of GG (Grade Group) 1. Hence risk-stratified pathways.
- **Biomarkers** (reflex tests before biopsy):

Test	Sample	Content
PHI (Prostate Health Index)	Blood	(p2PSA ÷ free PSA) × √total PSA
4Kscore	Blood	Four kallikreins + clinical data; % risk of GG ≥2
STHLM3	Blood	4Kscore kallikreins + 232 risk SNPs (single-nucleotide polymorphisms) + clinical variables
PCA3 (prostate cancer antigen 3), SelectMDx	Urine after DRE (digital rectal examination)	RNA ratio (PCA3); DLX1 and HOXC6 (SelectMDx)
ExoDx IntelliScore	Urine, no DRE	Exosomal RNA of PCA3, ERG, SPDEF

- Tissue genomic classifiers (Decipher, Prolaris, Oncotype DX) refine risk for surveillance and adjuvant decisions.
- **MRI-first pathway:** mpMRI (multiparametric magnetic resonance imaging), scored by **PI-RADS (Prostate Imaging–Reporting and Data System)** 1–5, before biopsy.
  - **PROMIS (2017):** mpMRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; NPV (negative predictive value) 89%; 27% could avoid biopsy.
  - **PRECISION (2018):** MRI with targeted biopsy of PI-RADS ≥3 found more significant cancer (38% vs 26%); 28% avoided biopsy.
  - So PI-RADS ≥3 lesions are biopsied, while PI-RADS 1–2 with low PSA density may safely avoid biopsy.
- **Transperineal biopsy** under local anaesthesia: near-zero sepsis; MRI–ultrasound fusion targeting. **Micro-ultrasound** (PRI-MUS scoring) matches mpMRI in registry data.
- **PSMA (prostate-specific membrane antigen) PET (positron emission tomography): proPSMA:** accuracy 92% vs 65% for CT (computed tomography) plus bone scan (sensitivity 85%, specificity 98%). Used for high-risk staging, biochemical recurrence and selecting radioligand therapy.

2. Localised disease

- **AS (active surveillance):** standard for GG1; extended to low-volume favourable intermediate risk (GG2), with MRI and PSA density. **ProtecT (15 years):** prostate cancer mortality about 2–3% in monitoring, surgery and radiotherapy arms; more metastasis on monitoring, no survival difference.
- **Robotic RP (radical prostatectomy): Retzius-sparing** (Bocciardi) gives earlier continence (1 week to 3 months; equal by 12 months); single-port robot; frozen-section-

guided nerve sparing (NeuroSAFE).

- **Radiotherapy:**
  - Moderate hypofractionation **60 Gy in 20** (CHHiP) is standard.
  - **SBRT (stereotactic body radiotherapy) 36.25 Gy in 5:** PACE-B showed similar late toxicity at 2 years (chapter); its 5-year report (2024) showed non-inferior biochemical or clinical control. HYPO-RT-PC (42.7 Gy in 7) was also non-inferior.
  - **FLAME:** MRI-lesion boost improved biochemical PFS (progression-free survival) (HR [hazard ratio] 0.45). **Hydrogel rectal spacer** reduces rectal toxicity.
  - High-risk M0: **abiraterone added to ADT (androgen deprivation therapy) and RT (STAMPEDE):** MFS (metastasis-free survival) HR 0.53.
- **Focal therapy:** based on the **index lesion hypothesis** (the largest or highest-grade focus drives progression). Energies: HIFU (high-intensity focused ultrasound), cryotherapy, IRE (irreversible electroporation), focal laser, VTP (vascular-targeted photodynamic therapy, padeliporfin). **HEAT registry (1379 men):** failure-free survival 68% (intermediate risk); continence 90–95%. Still within trials or registries.

3. Recurrent and advanced disease

Setting	Advance	Evidence
High-risk BCR (biochemical recurrence)	PSMA-guided salvage; enzalutamide + leuprolide	**EMBARK:** MFS HR 0.42
mHSPC (metastatic hormone-sensitive)	ADT + ARPI (androgen receptor pathway inhibitor)	LATITUDE, TITAN, ENZAMET, ARCHES
	Triplet (ADT + docetaxel + ARPI)	**PEACE-1, ARASENS**
	Prostate RT, low volume	STAMPEDE arm H
	MDT (metastasis-directed therapy)	STOMP, ORIOLE
nmCRPC (non-metastatic castration-resistant)	Apalutamide, enzalutamide, darolutamide	SPARTAN, PROSPER, ARAMIS
HRR (homologous recombination repair)-mutated mCRPC	**PARP (poly-ADP-ribose polymerase) inhibitors**	**PROfound:** olaparib imaging PFS 7.4 vs 3.6 months; PROpel, MAGNITUDE, TALAPRO-2
PSMA-positive mCRPC	**^{177}Lu-PSMA-617**	**VISION:** OS (overall survival) 15.3 vs 11.3 months (HR 0.62); TheraP: PSA response 65% vs 37% vs cabazitaxel

- **ADT alone is no longer standard** for mHSPC; every fit man gets intensification.
- **Germline and somatic testing** for all metastatic disease; **pembrolizumab** for MSI-high (microsatellite instability-high) tumours.
- **Relugolix:** oral GnRH (gonadotropin-releasing hormone) antagonist; HERO showed better sustained castration than leuprolide.
- **Also know:** radium-223 (ALSYMPCA, OS HR 0.70); PSMAfore (lutetium before chemotherapy); AI in MRI reading and pathology.

Diagram to draw: MRI-first pathway: raised PSA → mpMRI → PI-RADS ≥3 targeted transperineal biopsy; PI-RADS 1–2 with low PSA density → surveillance.

Key points to remember:

- PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
- proPSMA: PSMA PET accuracy 92% vs 65%.
- ProtecT: no mortality difference at 15 years; AS is standard for GG1.
- VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

Q3a. Virtual cystoscopy

Type: SAQ (10 marks)

Definition and principle

- **Virtual cystoscopy (VC)** is a computer-generated, three-dimensional (3D) endoluminal view of the bladder. It is reconstructed from thin-section **CT (computed tomography)** or **MRI (magnetic resonance imaging)** data of a distended bladder.
- Software "flies through" the lumen using **surface or volume rendering**; multiplanar images also show wall thickness and extravesical spread.

Technique

Step	CT virtual cystoscopy	MR virtual cystoscopy
Distension	Air or CO2 through a catheter (negative contrast), or excretory-phase contrast (positive contrast, no catheter)	Urine-filled bladder; urine bright on T2
Acquisition	Thin slices, supine and prone (moves dependent urine off the posterior wall)	3D T2 or T1 contrast sequences
Drawback	Radiation; catheter for gas	Cost, time, motion artefact

Advantages

- Non-invasive; no anaesthesia.
- Sees where a scope cannot: through a **tight urethral stricture**, inside **diverticula**, beyond a large median lobe.
- With **CTU (CT urography)**, upper tracts and staging are imaged in one study (Chapter 5: high accuracy for bladder tumours if well distended).

Limitations

- Detects polypoid tumours ≥5 mm well; misses **small and flat lesions, including CIS (carcinoma in situ)**.
- No colour or mucosal detail, **no biopsy and no treatment**.
- False positives: trabeculation, clot, debris, folds of an underdistended bladder, prostatic impression.

Indications (selective)

- Unfit for, or refusing, cystoscopy; urethral stricture; diverticulum; bladder evaluation during CTU for haematuria.
- **Guideline position:** EAU (European Association of Urology) and AUA (American Urological Association) do not recommend VC in place of cystoscopy. Flexible cystoscopy remains the standard.

Related advances

- **Also know:** MRI **VI-RADS (Vesical Imaging–Reporting and Data System)** scores muscle invasion (Chapter 6); blue-light and narrow-band cystoscopy improve CIS detection (Chapter 8); virtual ureteroscopy uses the same principle.

Diagram to draw: thin-slice CT of a gas-distended bladder, then a 3D endoluminal fly-through showing a polypoid tumour.

Key points to remember:

- 3D endoluminal reconstruction from thin-section CT or MRI.
- Good for polypoid tumours ≥5 mm; misses flat lesions and CIS.
- No biopsy; does not replace cystoscopy.
- Useful beyond strictures and inside diverticula.

Q3b. 3D laparoscopy

Type: SAQ (10 marks)

Introduction

- Standard laparoscopy gives a **two-dimensional (2D)** image. Depth is judged from monocular cues (shadow, size, motion parallax, touch), which slows suturing.
- **Three-dimensional (3D) laparoscopy** restores **stereoscopic depth perception**, as in robotic surgery (Chapter 18: two-lens binocular systems give depth perception).

Principle

- Two lenses or two chip-on-tip sensors, a few millimetres apart, capture two images from slightly different angles, like the two eyes.
- The monitor shows both images; **polarised glasses** send one to each eye. The brain fuses them, and the **binocular disparity** is perceived as depth.
- **Also know:** active-shutter glasses; head-mounted displays; glasses-free (autostereoscopic) screens; 3D scopes at 0° and 30°, and with a deflectable tip; 3D 4K systems.

Advantages

- Better depth judgement and hand–eye coordination.
- **Shorter time for complex tasks**, especially intracorporeal suturing and knotting, with fewer errors.
- **Shorter learning curve**; novices gain more than experts.
- Gives robotic-type vision at much lower cost.

Disadvantages

- Visual strain, headache, dizziness or nausea in some users; glasses dim the image.
- Every team member needs glasses and a correct viewing distance and angle.
- Higher cost than 2D; 30° rotation needs care.
- No wristed instruments, tremor filtering or motion scaling: these come only with the robot.

Urological uses

- Reconstructive steps: **pyeloplasty**, vesicourethral anastomosis in laparoscopic radical prostatectomy, **partial nephrectomy** renorrhaphy, ureteric reimplantation, donor nephrectomy.

Evidence

- Randomised trials and meta-analyses in laboratory tasks and laparoscopic surgery show shorter operating and suturing times with 3D. Differences in complications and oncological outcomes are not established.

Feature	2D laparoscopy	3D laparoscopy	Robot
Depth perception	No	Yes	Yes (magnified)
Wristed instruments	No	No	Yes
Cost	Low	Moderate	High

Diagram to draw: two lenses at the scope tip, two images, polarised glasses, fused 3D image in the brain.

Key points to remember:

- Two lenses, two images, polarised glasses: binocular disparity gives depth.
- Main gain: faster, more accurate suturing; shorter learning curve.
- Drawbacks: eye strain, headache, cost, glasses for all.
- Robotic vision at low cost, but without wristed instruments.

Q3c. Telemedicine in urology

Type: SAQ (10 marks)

Definition

- **Telemedicine** is delivery of health care at a distance using information and communication technology (video, phone, messaging, data transfer).
- **COVID-19 (coronavirus disease 2019)** distancing accelerated its adoption. Chapter 2: urologists' telehealth use rose sharply in 2019–2020.

Types

Type	Method	Urological example
Synchronous	Real-time video or phone consultation	Stone, PSA (prostate-specific antigen) and post-operative follow-up
Asynchronous (store-and-forward)	Reports, images, bladder diaries sent for later review	Scan reports, uroflow traces
Remote monitoring	Apps and home devices	Bladder diary apps, home uroflowmetry
Provider-to-provider	E-consults, telementoring, tele-tumour boards	Haematuria triage (Chapter 8)
Telesurgery	Remote robotic surgery	Experimental; limited by latency

Uses in urology

- **History-driven care:** LUTS (lower urinary tract symptoms) review, results, drug titration.
- ED (erectile dysfunction) evaluation by selected telemedicine (Chapter 72); men's health in large, sparse regions (Chapter 68).
- After mid-urethral sling, telemedicine follow-up is safe, and app-based follow-up gave higher satisfaction than office visits (Chapter 116).
- Internet-based pelvic floor training equalled a postal programme in an RCT (randomised controlled trial) (Chapter 114).

Benefits (Chapter 2)

- Access for rural and underserved patients; less travel; shorter waits; relatives join remotely; sustained satisfaction.

Limitations (Chapter 2)

- **No examination or in-office tests**: vital signs, cystoscopy, PVR (post-void residual).
- Needs connectivity and a secure, simple platform; hard for elderly or non-technical patients.
- No assistant to pre-gather results; more no-shows.
- **Medico-legal:** US licensure is state-based, and pandemic payment waivers have ended.

Indian framework

- **Telemedicine Practice Guidelines (2020)**: only a registered medical practitioner may consult; identity of patient and doctor confirmed; consent and records kept; narcotic and Schedule X drugs may not be prescribed.
- Government platform **eSanjeevani**; patient data covered by the Digital Personal Data Protection Act 2023.

Diagram to draw: patient (home or district hospital) linked by video and data to the urologist, with remote monitoring feeding in.

Key points to remember:

- Telemedicine trades examination for access.
- Best for follow-up, results and history-driven visits.
- Limits: no examination, connectivity, elderly, medico-legal.
- India: Telemedicine Practice Guidelines 2020; eSanjeevani.

Q3d. Simulators in urology

Type: SAQ (10 marks)

Rationale

- Fewer training cases and complex technology mean skills must be learnt **outside the patient**.
- Chapter 1: simulation is used to **acquire and maintain skills and to warm up** before operating.

Types

Type	Examples	Strengths and limits
Bench / box trainer	Peg transfer, cutting, suturing; chicken oesophagus anastomosis; 3D (three-dimensional)-printed perfused hydrogel kidney	Cheap, proven skill gains; limited realism
VR (virtual reality)	LapSim, LapMentor, RobotiX Mentor, SimNow (robotic prostatectomy); endourology trainers for cystoscopy, ureteroscopy, PCNL (percutaneous nephrolithotomy), TURP (transurethral resection of the prostate)	Records metrics (precision, economy of motion), builds curricula, sets expert benchmarks; costly; poor tissue feel
Live animal	Pig	Real bleeding; ethics, cost
Cadaver	Fresh-frozen	Real human anatomy; scarce
Team (non-technical)	High-fidelity theatre crises	Communication and teamwork

Validity (needed before a simulator is trusted)

- **Face:** looks real. **Content:** teaches the right skills. **Construct:** experts score better than novices. **Concurrent:** agrees with an established test. **Predictive:** simulator score forecasts operating performance.

Evidence (Chapter 18, Chapter 12)

- VR training **beats no training or box training** for novices and improves live performance and operating time.
- Robotic simulator scores correlate with live RARP (robot-assisted radical prostatectomy) performance.
- **FRS (Fundamentals of Robotic Surgery):** a multi-institutional RCT (randomised controlled trial) showed better speed and accuracy.
- Ureteroscopy team simulation taught non-technical skills better than lectures (randomised trial).

Structured curricula and assessment

- **EAU (European Association of Urology) E-BLUS** (European Basic Laparoscopic Urological Skills): online theory, then dry-model tasks, then examination.
- **Proficiency-based progression:** move on only after reaching expert benchmarks.
- Scoring: **GEARS (Global Evaluative Assessment of Robotic Skills)** and **OSATS (Objective Structured Assessment of Technical Skills)**.

Limitations

- Cost, poor tissue feel, limited proof of better patient outcomes.

Diagram to draw: pyramid: box trainer at the base, then VR, then animal or cadaver, then supervised surgery at the top.

Key points to remember:

- Bench, VR, animal, cadaver and team simulation.
- Validity: face, content, construct, concurrent, predictive.
- VR beats box or no training for novices.
- E-BLUS and FRS; proficiency-based progression; GEARS.

Q3e. Virtual and augmented reality in urology

Type: SAQ (10 marks)

Definitions

- **Virtual reality (VR):** the user is immersed in a fully computer-generated environment, usually through a head-mounted display.
- **Augmented reality (AR):** digital images are laid over the real view, for example a 3D (three-dimensional) tumour model on the robotic camera image.
- **Mixed reality:** virtual holograms are anchored in the room and can be handled.

Workflow

```
CT (computed tomography) / MRI (magnetic resonance imaging)
   -> segmentation of organ, tumour, vessels, collecting system
   -> 3D virtual model -> viewed in VR, 3D-printed, or overlaid (AR)
   -> registration to the live surgical view
```

Applications

Use	Examples
Pre-operative planning	Partial nephrectomy: tumour depth, arterial branches, selective clamping. **PCNL (percutaneous nephrolithotomy)**: calyx and tract choice (Chapter 98: immersive VR and 3D printing show early promise). Prostatectomy: MRI lesion model to plan nerve-sparing
Intra-operative navigation	AR overlay in the robot console to locate an endophytic tumour, hilar vessels, or a prostate lesion near the capsule
Education	VR anatomy, 360° operation videos; simulation, see Q3d
Patient counselling	Patients understand their anatomy and planned operation; better-informed consent
Paediatrics	VR distraction reduces anxiety before surgery and procedures such as catheterisation and VCUG (voiding cystourethrogram) (Chapters 35 and 38)

Limitations

- **Registration error:** organs move and deform with breathing, pneumoperitoneum and dissection, so a rigid overlay drifts; manual alignment is still common.
- Segmentation time and cost.
- Chapter 161: promising but **resource-intensive and hard to validate**.
- Chapter 98: 3D reformatting added nothing over orthogonal CT measurements for predicting stone-free rate.
- Little outcome evidence; cybersickness with headsets.

Future

- **AI (artificial intelligence)** for automatic segmentation and real-time tracking.
- **Digital surgery:** reconstruction of the patient's own anatomy allows rehearsal before the operation (Chapter 147).

Diagram to draw: CT kidney, segmented 3D model, and overlay of the model on the robotic view of a renal tumour.

Key points to remember:

- VR immerses; AR overlays on reality.
- Main uses: planning partial nephrectomy, PCNL and prostatectomy; navigation; teaching; counselling.
- Biggest problem: registration on moving, deforming organs.
- Evidence for outcomes still limited.

Q3f. Bladder transplantation

Type: SAQ (10 marks)

Rationale

- After cystectomy, or when the bladder cannot be reconstructed, the usual substitute is **bowel**. Bowel in the urinary tract causes mucus, metabolic acidosis, stones, infection and malignancy risk.
- Chapter 135: two frontiers aim to **remove bowel from urinary reconstruction**: tissue engineering and bladder transplantation.

Tissue engineering (why it has not succeeded)

- **Atala et al. (2006):** first seeded biodegradable construct for neurogenic augmentation. A later multi-institution trial showed **no significant gain in compliance or capacity** (Chapter 48).
- Engineered conduit trial: **half developed stenosis or stricture**, because of stiff material, poor cell coverage and **no blood supply** (Chapter 135).
- Engineered bladder: storage only (Chapter 111). See 2022-Summer Paper 4, Q2.

Bladder transplantation: development (Chapter 135)

- A **VCA (vascularised composite allograft)**: the donor bladder is transplanted with its own blood vessels.
- Stepwise preclinical work at USC (University of Southern California): pigs, perfused cadavers, then brain-dead human donors.

Technique (robotic)

1. **Robotic retrieval** of the bladder on its vesical vascular pedicles from the donor.
2. **Back-table vascular reconstruction**.
3. **Robotic implantation**: vessels to recipient iliac vessels; ureters joined to the graft.
4. **ICG (indocyanine green) fluorescence** confirms perfusion. Vascularity and mucosal integrity held for over 12 hours.

First clinical case

- Reported in **2025** from Los Angeles: a combined kidney and bladder transplant from a deceased donor, in a dialysis patient who had lost most of his bladder.

Problems

- **Lifelong immunosuppression**: infection, malignancy, drug nephrotoxicity. This is the major limitation (Chapter 135), especially after cancer.
- **Denervated graft**: stores urine; voiding may need straining or CIC (clean intermittent catheterisation).
- Rejection monitoring, donor supply, unknown long-term function.

Likely candidates

- Already immunosuppressed patients (kidney transplant recipients) with an absent or non-functioning bladder.

Diagram to draw: donor bladder on its vascular pedicles anastomosed to recipient iliac vessels, ureters entering the graft.

Key points to remember:

- Goal: avoid bowel in the urinary tract.
- Tissue-engineered bladders failed on blood supply and function.
- Robotic VCA; first human case 2025.
- Limits: immunosuppression and denervation.

Rapid Revision Sheet – Practice Set 5 Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- proPSMA: PSMA PET accuracy 92% vs 65%.
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- VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

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- 3D endoluminal reconstruction from thin-section CT or MRI.
- Good for polypoid tumours ≥5 mm; misses flat lesions and CIS.
- No biopsy; does not replace cystoscopy.
- Useful beyond strictures and inside diverticula.

Q3b. 3D laparoscopy (SAQ)

- Two lenses, two images, polarised glasses: binocular disparity gives depth.
- Main gain: faster, more accurate suturing; shorter learning curve.
- Drawbacks: eye strain, headache, cost, glasses for all.
- Robotic vision at low cost, but without wristed instruments.

Q3c. Telemedicine in urology (SAQ)

- Telemedicine trades examination for access.
- Best for follow-up, results and history-driven visits.
- Limits: no examination, connectivity, elderly, medico-legal.
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Q3d. Simulators in urology (SAQ)

- Bench, VR, animal, cadaver and team simulation.
- Validity: face, content, construct, concurrent, predictive.
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- Limits: immunosuppression and denervation.